

Maven POM.xml – Complete Interview Notes (Final)

1) What is POM.xml?

POM stands for Project Object Model. It is the core configuration file in Maven used to define project details, dependencies, plugins, and build configuration.

2) Dependency in POM.xml

Dependencies are external libraries required for the project.

```
<dependency>
  <groupId>org.springframework</groupId>
  <artifactId>spring-core</artifactId>
  <version>5.3.0</version>
</dependency>
```

3) Packaging Concept

When we build the project, Maven creates a package like JAR or WAR.

artifactId-version.jar (Example: myapp-1.0.jar)

4) Why Unique Naming?

Project names must be globally unique, so we use reverse domain naming.

5) groupId

Represents organization name in reverse domain format.

Example: com.company.project

6) artifactId

Represents project/module name.

Example: student-management-system

7) version

Represents project version.

Examples: 1.0, 1.0-SNAPSHOT

8) Complete Example

```
<project>
  <groupId>com.company</groupId>
  <artifactId>myapp</artifactId>
  <version>1.0</version>
</project>
```

9) Refresh in Eclipse

Right click project → Maven → Update Project → OK
Shortcut: Alt + F5

10) Effective POM

Effective POM is the final POM used by Maven after combining:

- Project POM • Parent POM • Super POM

Why it is used?

- To understand final configuration • Debug dependency issues • View inherited configurations

Important Point

Effective POM is created internally for every Maven project.

How to check in Eclipse?

Method 1:

Open pom.xml → Bottom tabs → Click 'Effective POM'

Method 2:

Right click project → Maven → Show Effective POM (if available)

Important Note (Your Case)

Even if 'Show Effective POM' is not visible in right-click menu, you can always see it by opening pom.xml and selecting the 'Effective POM' tab at the bottom.

11) Interview Q&A;

Question	Answer
What is POM?	Project Object Model
groupId?	Organization name
artifactId?	Project name
version?	Project version
Effective POM?	Final merged POM

12) Final Summary

POM.xml is the heart of Maven. It manages dependencies, defines project identity, and controls build. Effective POM shows the final configuration after merging all sources.